

UjenziXform TIS — Installation & Deployment Guide

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Status: Applicant / in development — production eTIMS cutover only after KRA certification.

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1. Scope

This guide covers installation and deployment of the **UjenziXform Trader Invoicing System (TIS)** — the eTIMS integration layer within the UjenziXform platform.

It is intended for:

- UjenziXform technical staff deploying sandbox or production environments
- KRA reviewers verifying how TIS software is installed and configured

This guide does **not** cover general UjenziXform marketplace features unrelated to eTIMS (payments, delivery, messaging).

2. System requirements

2.1 Runtime stack

Layer	Technology	Minimum version
Frontend	React, TypeScript, Vite	Node.js 18+
Backend	Supabase (PostgreSQL, Auth, RLS)	Supabase project
API gateway	Supabase Edge Functions (Deno)	etims-proxy deployed
Hosting	Vercel or equivalent static host	HTTPS required

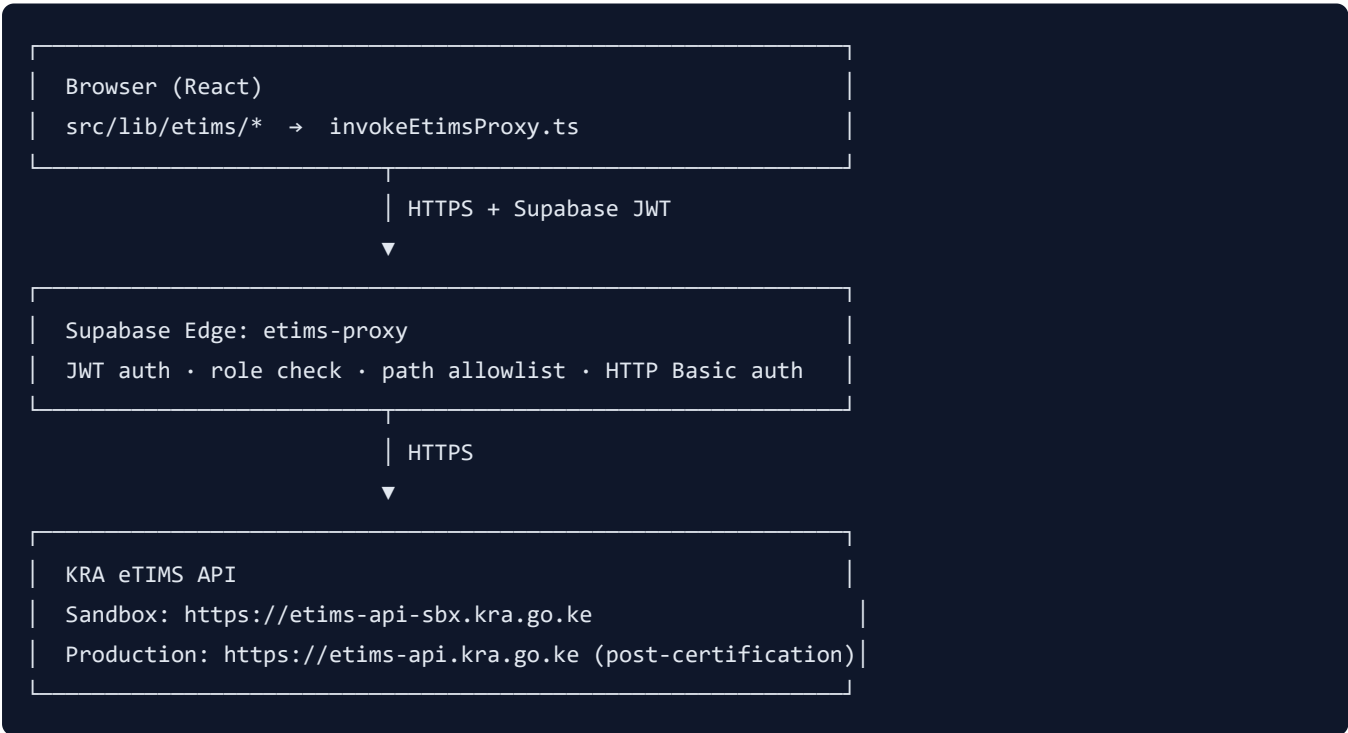
2.2 External services

Service	Purpose
KRA eTIMS API	Fiscal invoice submission (sandbox or production)
Supabase Dashboard	Edge secrets, migrations, auth

2.3 Operator access

Role	Required for
Supabase project admin	Deploy Edge functions, set secrets, run migrations
Vercel (or host) admin	Deploy frontend, set public env vars
eTIMS portal account	Integrator registration and sandbox credential request

3. Architecture overview



KRA credentials (`ETIMS_BASIC_USER` , `ETIMS_BASIC_PASSWORD`) exist **only** in Supabase Edge secrets. They are never committed to git, never stored in PostgreSQL, and never exposed to the browser.

4. Software components

4.1 TIS integration source (included on optical disk)

Path	Description
<code>src/lib/etims/</code>	Core TIS library (PO mapping, OSCU init, audit)
<code>src/components/admin/tis-integrator/</code>	Admin TIS Integrator Hub
<code>src/components/etims/</code>	Fiscal receipt UI
<code>src/components/admin/Etims*.tsx</code>	Admin test and PO submit panels
<code>src/components/supplier/SupplierEtimsSettingsPanel.tsx</code>	Supplier KRA settings
<code>supabase/functions/etims-proxy/</code>	Secure KRA API gateway
<code>supabase/functions/_shared/etimsPathAllowlist.ts</code>	SSRF path allowlist
<code>supabase/migrations/etims / tis</code>	Database schema for TIS

4.2 Key entry points

Function	File
Build SalesReq from PO	<code>src/lib/etims/purchaseOrderEtims.ts</code>
Submit PO to eTIMS	<code>submitEtimsInvoiceForPurchaseOrder()</code>
OSCU initialization	<code>src/lib/etims/tisOscuInitialization.ts</code>
Audit logging	<code>src/lib/etims/logTisSubmission.ts</code>
Proxy invocation	<code>src/lib/etims/invokeEtimsProxy.ts</code>

5. Development environment setup

5.1 Clone and install

```
git clone <repository-url>
cd UjenziXform
npm install
```

5.2 Environment file

Copy the template and configure public variables only:

```
cp env.local.template .env.local
```

Variable	Required	Notes
VITE_SUPABASE_URL	Yes	Supabase project URL
VITE_SUPABASE_ANON_KEY	Yes	Public anon key — safe for browser

Do not put `ETIMS_BASIC_USER` or `ETIMS_BASIC_PASSWORD` in `.env.local`. These belong in Supabase Edge secrets only.

5.3 Start development server

```
npm run dev
```

Default Vite dev server: `http://localhost:5173`

5.4 Run tests

```
npm run test:run
```

TIS-related unit tests include `purchaseOrderEtims.test.ts`, `formatEtimsReceiptForUi.test.ts`, and `kraPin.test.ts`.

6. Database migrations

TIS requires PostgreSQL tables and RLS policies created by Supabase migrations.

6.1 Key migrations

Migration	Purpose
20260516180000_kra_etims_profiles_suppliers_products.sql	Supplier/buyer tax fields, item codes
20260525120000_tis_integrator_platform.sql	Integrator platform metadata, checklist
20260525130000_fix_tis_integrator_rls_roles.sql	RLS for TIS admin tables
20260525140000_tis_onboarding_init_audit.sql	Vendor onboarding, init audit
PO <code>etims_*</code> column migrations	Fiscal outcome on purchase orders

6.2 Apply migrations

Option A — Supabase CLI (linked project):

```
supabase db push
```

Option B — Supabase Dashboard:

1. Open **SQL Editor**.
2. Run migration SQL files in chronological order from `supabase/migrations/`.

6.3 Verify tables

Confirm these tables exist:

- `tis_integrator_platform`
- `tis_vendor_onboarding`
- `tis_submission_log`
- `purchase_orders` with columns `etims_validated_at`, `etims_response`, `etims_error`

7. Supabase Edge Function deployment

7.1 Deploy etims-proxy

```
supabase functions deploy etims-proxy
```

7.2 Function behaviour

Control	Implementation
HTTP method	POST only (OPTIONS for CORS)
Authentication	Valid Supabase JWT required
Authorization	Roles: <code>supplier</code> , <code>admin</code> , <code>super_admin</code>
Path validation	Allowlist in <code>etimsPathAllowlist.ts</code>
Body size limit	400 KB maximum
Upstream auth	HTTP Basic using Edge secrets

7.3 Invoke format

```
{
  "method": "POST",
  "path": "invoices",
  "body": { "... SalesReq JSON ..." }
}
```

The Edge function calls `{ETIMS_BASE_URL}/{path}` with the configured Basic credentials.

8. eTIMS / KRA credential configuration

8.1 Edge secrets (Supabase Dashboard)

Navigate to: **Project Settings** → **Edge Functions** → **Secrets**

Secret	Sandbox value (after KRA assignment)	Production value
<code>ETIMS_BASE_URL</code>	<code>https://etims-api-sbx.kra.go.ke</code>	<code>https://etims-api.kra.go.ke</code>
<code>ETIMS_BASIC_USER</code>	Issued by KRA	Issued by KRA
<code>ETIMS_BASIC_PASSWORD</code>	Issued by KRA	Issued by KRA

Rules:

- No trailing slash on `ETIMS_BASE_URL`
- URL must start with `http://` or `https://`
- Rotate credentials if compromised; never log passwords

8.2 Platform certification status

In the admin **TIS Integrator Hub** → **Platform & certification** panel, set:

Field	Sandbox testing
Environment	<code>sandbox_testing</code>
Base URL reference	<code>https://etims-api-sbx.kra.go.ke</code>

Update to `certified` and production URL only after formal KRA approval.

8.3 Verify configuration

1. Sign in as admin.
2. Open Admin dashboard → **eTIMS test** tab.
3. Run a connection test or invoke a safe GET reference-data path.
4. Expected: HTTP 200 from KRA sandbox (not 503 “proxy not configured”).

9. Frontend deployment

9.1 Build

```
npm run build
```

Output directory: `dist/`

9.2 Host environment variables

Set on Vercel (or equivalent) for **Production** and **Preview**:

Variable	Value
<code>VITE_SUPABASE_URL</code>	Supabase project URL
<code>VITE_SUPABASE_ANON_KEY</code>	Supabase anon key

Redeploy after changing environment variables.

9.3 CORS

The `etims-proxy` Edge function returns CORS headers for browser invocation. Ensure the Supabase project URL matches the deployed frontend’s configured Supabase client.

10. Post-deployment verification

Run this checklist after each deployment to a new environment:

#	Check	Pass criteria
1	Edge secrets set	No 503 from <code>etims-proxy</code>
2	Migrations applied	TIS tables queryable
3	Admin TIS Integrator Hub loads	Four sub-tabs visible
4	Supplier KRA settings save	<code>suppliers.kra_pin</code> persisted
5	Test PO submit (sandbox)	<code>etims_validated_at</code> set; receipt displays QR
6	Audit log	Row in <code>tis_submission_log</code>
7	Secrets scan	No credentials in git or optical disk bundle

11. Sandbox vs production environments

Aspect	Sandbox	Production
KRA base URL	<code>https://etims-api-sbx.kra.go.ke</code>	<code>https://etims-api.kra.go.ke</code>
Credentials	KRA sandbox Basic auth	KRA production Basic auth
Device serials	KRA sandbox test devices	Live OSCU devices
Go-live gate	Internal test plan complete	KRA integrator certification
Platform status field	<code>sandbox_testing</code>	<code>certified</code>

Production cutover procedure:

1. Obtain KRA production credentials and certification approval.
2. Update Edge secrets to production URL and credentials.
3. Re-run OSCU initialization for each active vendor device on production.
4. Update `tis_integrator_platform.certification_status` to `certified`.
5. Monitor `tis_submission_log` for 48 hours post-cutover.

12. Security checklist

#	Control	Verification
1	KRA credentials in Edge secrets only	Grep repo — no passwords in source
2	JWT on every proxy call	Unauthenticated requests return 401
3	Role-based access	Non-supplier/admin roles blocked
4	Path allowlist	Arbitrary URLs rejected by proxy
5	RLS on TIS tables	Non-staff cannot read integrator platform
6	Communication keys redacted	Audit logs mask <code>cmcKey</code>
7	Optical disk has no <code>.env</code>	Run bundle script; manual review before burn

Run optional security audit:

```
npm run security:test
```

13. Optical disk software bundle

Before submitting the read-only CD/DVD to KRA, generate the software folder:

```
powershell -File scripts/build-tis-optical-disk.ps1
```

Output:

```
downloads/TIS-OPTICAL-DISK/
├─ README.txt
├─ SOFTWARE_MANIFEST.txt
├─ 01-SOFTWARE/      ← TIS source files (50+ files)
├─ 02-PDF-DOCUMENTS/ ← add exported PDFs manually
```

Review `SOFTWARE_MANIFEST.txt` for completeness. Confirm no secrets before burning.

14. Document control

Version	Date	Changes
1.0	2026-06-03	Initial release for KRA optical disk submission

Related documents:

- `TIS_ETIMS_INTEGRATION_ARCHITECTURE.md` — architecture (§14, §21)
- `USER_OPERATOR_MANUAL.md` — operator procedures
- `CERTIFICATION_TEST_PLAN.md` — sandbox test cases
- `OPTICAL_DISK_SUBMISSION_GUIDE.md` — disk assembly instructions